Computer Parts

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The Motherboard (bare motherboard)

As the core connectivity platform of a computer system, the motherboard integrates and connects various hardware components, including the CPU, memory, hard drive, and graphics card. It also transmits electronic signals and power, provides power management and data coordination, and supports additional peripheral devices and functions through expansion slots. The motherboard's design directly determines the stability, performance ceiling, and expansion capabilities of the entire computer system.



The CPU (CPU, Socket)

The Central Processing Unit (CPU) functions as the computer's "brain," responsible for executing program instructions to perform arithmetic, logical, and input/output operations, thereby transforming data input into usable output. The CPU socket is the specific slot on the motherboard where the CPU is installed. It has a lever or latch that opens to allow placement of the CPU and closes to secure it firmly. The socket has notches or markers to help align the CPU correctly, ensuring the pins or contacts match perfectly for proper connection and functionality.





How to Install CPU (cpu notches and markings etc, cpu in socket)

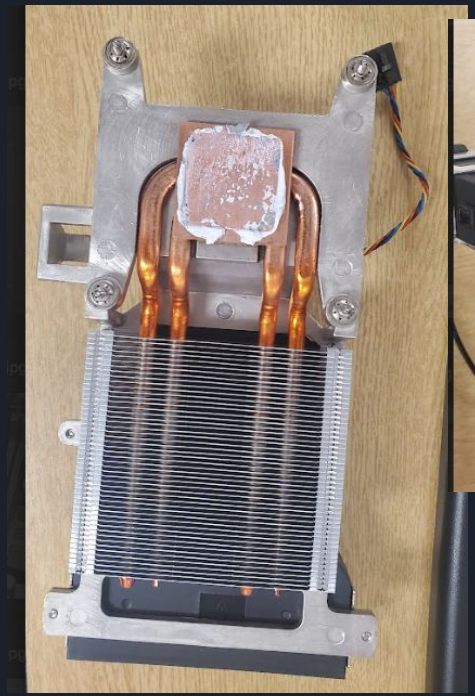
To install a CPU, first power off your computer and unplug all cables. Open the case and lift the CPU socket lever. Carefully align the CPU with the socket using the notches or triangle mark, then gently place it in without forcing. Close the socket lever to secure the CPU. Apply a small amount of thermal paste on the CPU surface, then install and secure the CPU cooler. Connect the cooler’s fan cable to the motherboard, close the case, reconnect cables, and power on the system to check everything works.

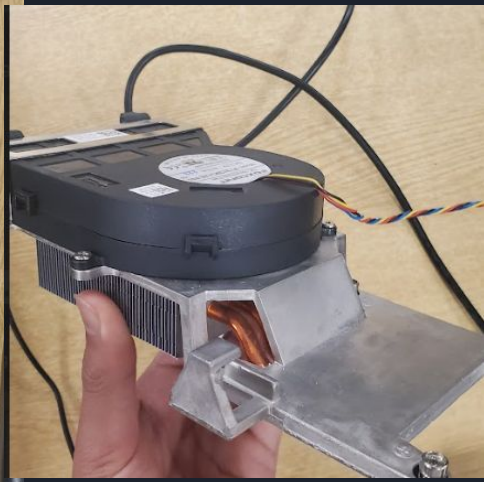




CPU Cooler CPU heatsink and fan combo, CPU heatsink

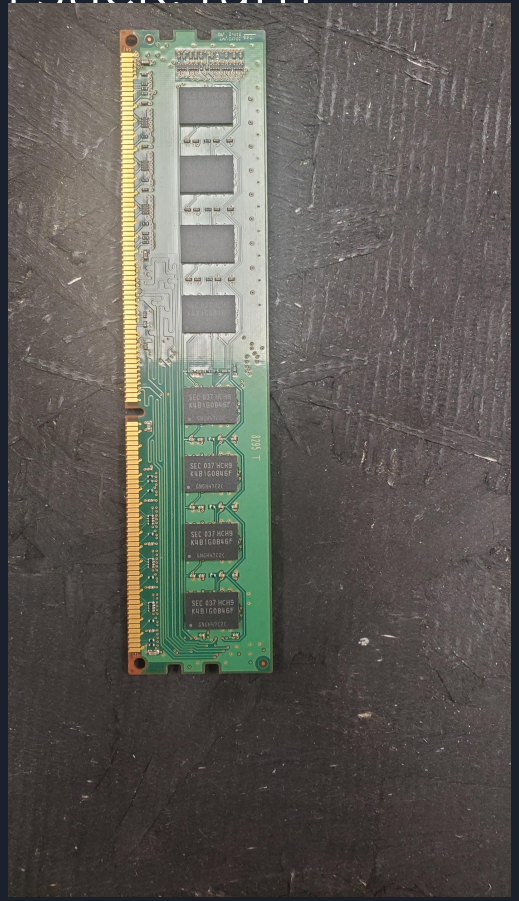
The CPU heatsink uses copper pipes and heat-conductive thermal paste to transfer away the heat from the CPU. This is paired with a fan which helps blow away the hot air.





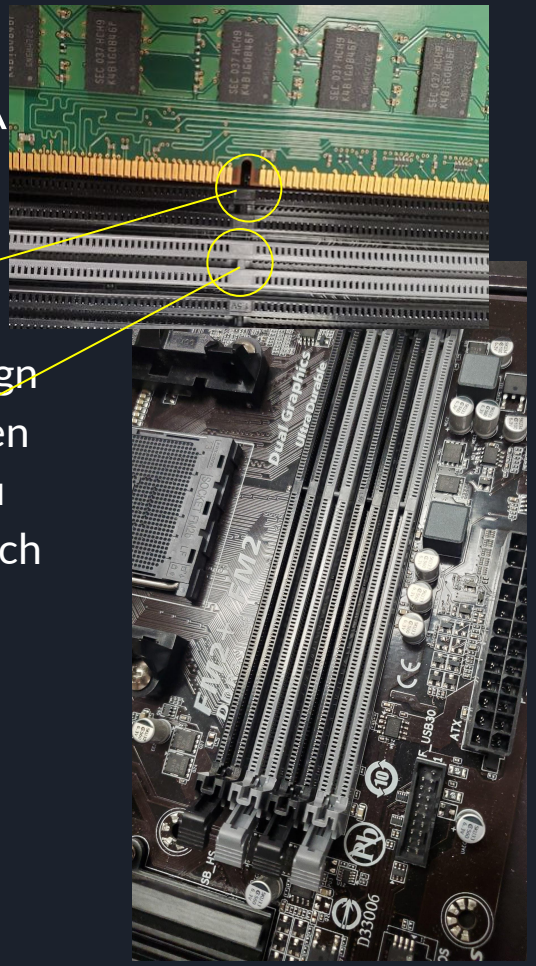
RAM (Random Access Memory) (ram stick, ram slot, notches/markings, ram in slot)

RAM, otherwise known as Random Access Memory, is your computer's temporary, high-speed working memory that the CPU needs to quickly access for running programs and processing tasks. RAM loses its data once the computer shuts down or loses power.



How to install RA

Unclip the RAM slot clips, make sure to align the RAM with the RAM slot connector, then insert the RAM in and push down until you hear a click or when the RAM slot clips, latch the RAM into place. Congratulations, you have installed RAM into your computer.



Motherboard Heatsinks (heatsink 1+2)

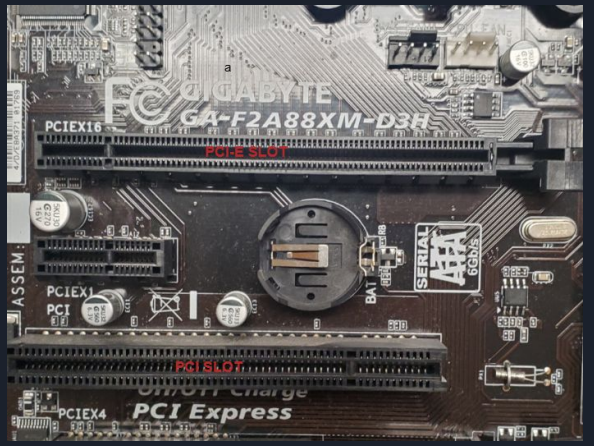
The Heatsinks on the motherboard help cool down the Northbridge and Southbridge chips. The Northbridge chip handles high-speed communication with the CPU and RAM while the Southbridge chip handles slower communication with storage devices and I/O devices.

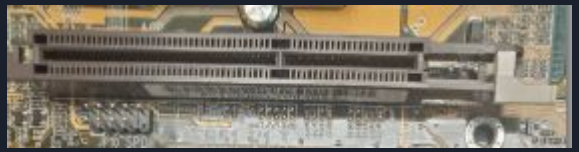


PCI and AGP slots (PCIE, PCI, AGp slots)

The PCI slot is a motherboard connector that connects expansion cards (video, audio, capture, etc). The PCI-E slot is a newer version that uses high-speed serial connections for more modern video and expansion cards.

The AGP (Accelerated Graphics Port) slot is similar to the PCI slot, but it's faster and allowed for more performance from video cards. The AGP lost popularity as the newer PCI-E slot took over.

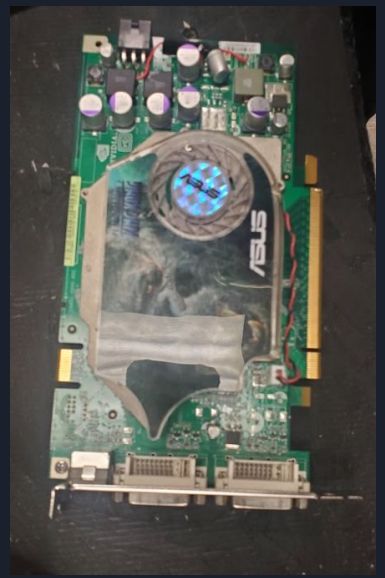


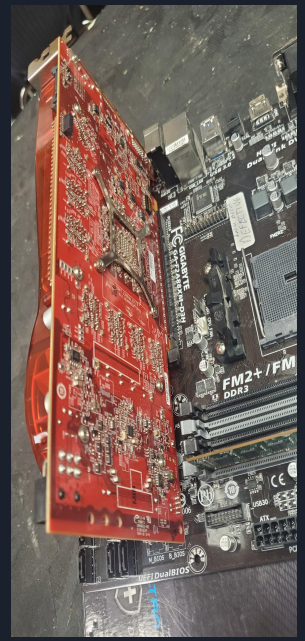


Video Card (GPU)

A GPU (graphics processing unit) is an expansion card designed to handle graphics and video-related rendering and modeling. They are generally used in gaming, video editing, and artificial intelligence.

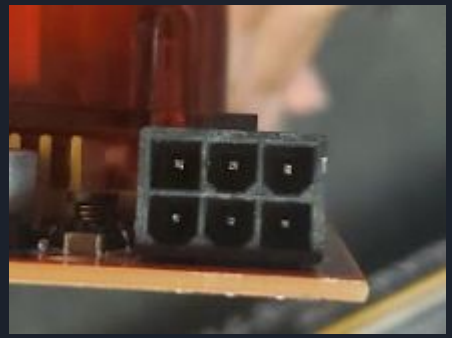
A GPU would generally fit into a PCI-E slot and uses anywhere from 6-16 pins (depending on the GPU's performance) from a 6+2 power connector.





Ex: 6+2 connector





Audio Card

An **Audio Card** (also called a sound card) is an expansion card or built-in component in a computer that processes audio signals. It converts digital audio data into sound that can be played through speakers or headphones and captures audio from microphones. Audio cards improve sound quality and provide features like surround sound, multiple audio inputs/outputs, and advanced audio processing compared to basic onboard audio. They connect to the motherboard via PCI or PCIe slots or are integrated directly on the motherboard.



Hard Disk Drive (Storage) (2.5+3.5 inch hdd)



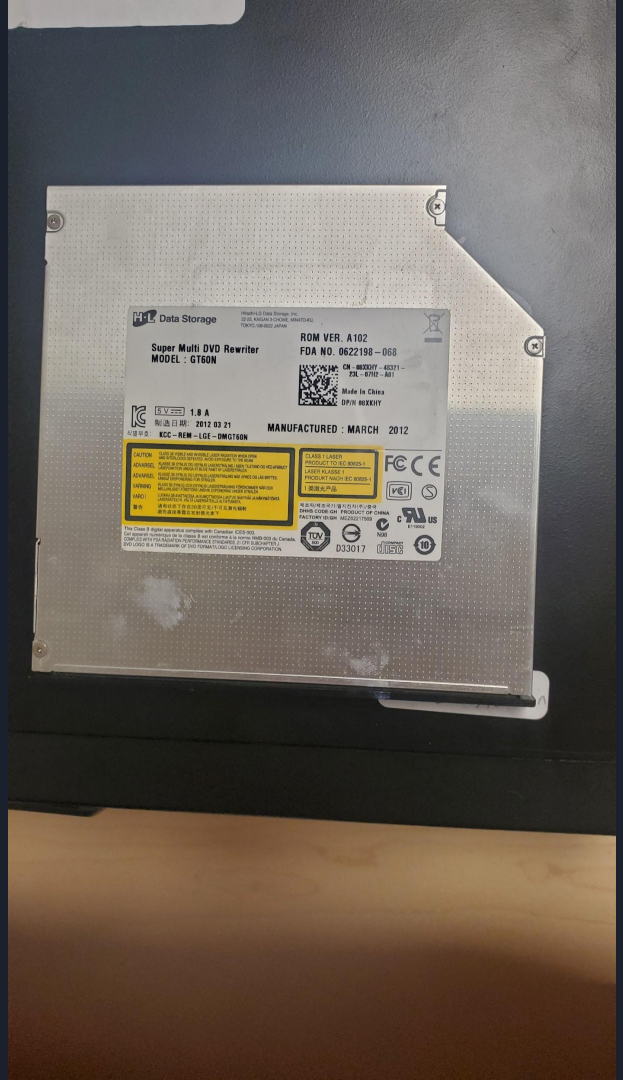
A **Hard Disk Drive (HDD)** is a storage device used to store data permanently in a computer. It uses spinning magnetic disks to read and write data. HDDs come in different sizes:

- **3.5-inch HDD:** Typically used in desktop computers, larger in size and capacity.
- **2.5-inch HDD:** Smaller size, commonly used in laptops or external drives.

Both types connect to the motherboard via SATA data cables and to the power supply via SATA power cables. HDDs are used to store the operating system, software, files, and other data.

ODD (Optical Disk Drive)

ODD (Optical Disk Drive) is a hardware device used to read and/or write data from optical discs like CDs, DVDs, or Blu-ray discs. It connects to the motherboard via SATA cables for data transfer and to the power supply for power. ODDs are commonly used for installing software, playing media, or backing up data on physical discs, although they are less common now due to digital downloads and streaming.



Sata Cable and Header (SATA data+power cable, SATA mb header)





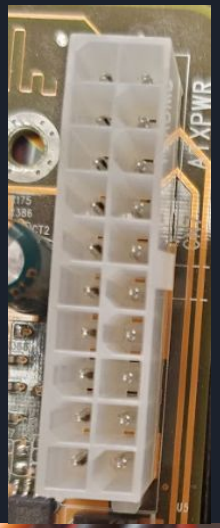
SATA Cable and Header:

- **SATA Data Cable:** This is a thin, flat cable used to connect storage devices like SSDs or HDDs to the motherboard. It transmits data between the drive and the motherboard.
- **SATA Power Cable:** This cable comes from the power supply unit (PSU) and provides power to the SATA storage devices.
- **SATA Motherboard Header:** These are the ports on the motherboard where the SATA data cables plug in. Each header connects to one SATA device.



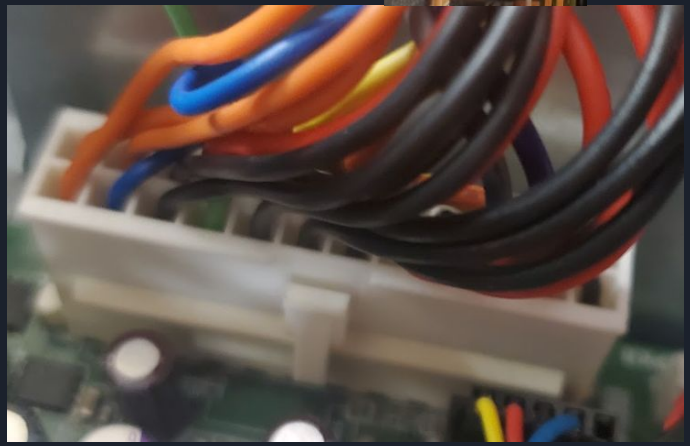
Together, the SATA data cable and power cable enable the storage device to communicate with the computer and receive power to operate.

20/24 Pin Motherboard Connector



The **20/24 Pin Motherboard Connector** is the main power connector from the power supply to the motherboard. It delivers the necessary power to the motherboard and all its components.

- **20-pin connector:** Older standard, used in earlier motherboards.
- **24-pin connector:** Modern standard, provides more power and supports newer motherboards with additional features.



Many power supplies have a 20+4 pin design, allowing compatibility with both types. This connector is essential for the motherboard to function properly.

4/8 Pin Motherboard Header

The **4/8 Pin Motherboard Header** (often called the CPU power connector) is a power socket on the motherboard that supplies additional, stable power specifically to the CPU. It usually comes in two forms:

- **4-pin connector:** Provides basic power for the CPU, common in older or less power-demanding systems.
- **8-pin connector (EPS12V):** Provides more power and is common in modern motherboards and high-performance CPUs.



The power supply unit (PSU) has matching 4-pin or 8-pin cables that plug into this header to ensure the CPU gets enough clean and stable power for proper operation.

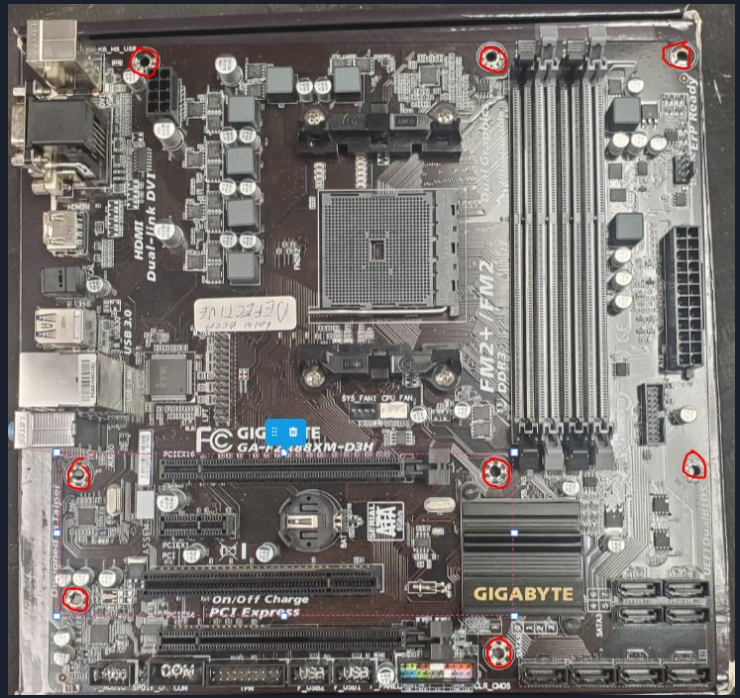
The PSU

The **PSU (Power Supply Unit)** is a critical component in a computer that converts the electrical power from an outlet into usable power for the computer's internal components. It supplies stable and regulated voltage to parts like the motherboard, CPU, GPU, drives, and fans. The PSU connects to the motherboard and other hardware using various power cables and connectors, and it often includes a fan to keep it cool. A good PSU ensures your system runs reliably and protects components from power surges or instability.



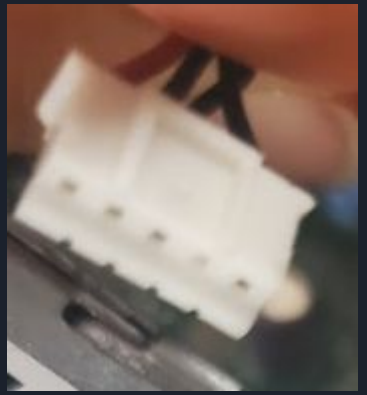
Motherboard Standoff holes

Motherboard Standoff Holes are the small holes on a motherboard designed to align with the standoffs (spacers) in a computer case. These holes allow the motherboard to be securely mounted to the case without directly touching the metal surface, preventing electrical shorts and providing stable support. The standoffs screw into the case, and the motherboard is then fixed onto these standoffs using screws through these holes. Proper alignment of the motherboard standoff holes with the case standoffs is essential for safe and secure installation.



Motherboard Fan connector

The **Motherboard Fan Connector** is a small socket on the motherboard designed to connect cooling fans, such as the CPU fan or case fans. It supplies power to the fan and sometimes allows the motherboard to control the fan speed based on temperature sensors. These connectors usually come in 3-pin or 4-pin varieties, where the extra pin allows for PWM (Pulse Width Modulation) speed control. Properly connecting fans to these headers helps keep your system cool and running efficiently.





The CMOS battery (cmos battery bare + in holder)

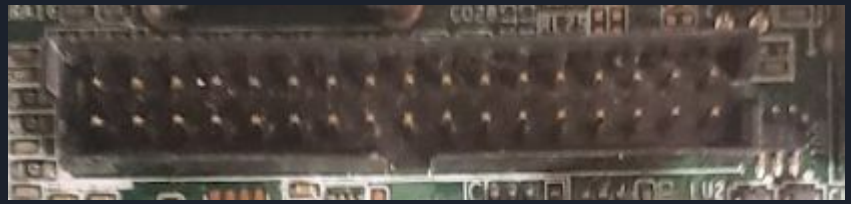
The CMOS battery is a small, round battery (usually a coin cell) on the motherboard that powers the CMOS memory, which stores BIOS settings like system time and hardware configurations when the computer is turned off. The battery sits inside a **battery holder**—a circular slot or clip on the motherboard that securely holds the battery in place while allowing easy removal or replacement. The battery is typically labeled with a “+” sign indicating the positive side, which should face up when installed in the holder.





Front Panel IO ports (header, connector, connector plugged in)

Front Panel IO ports refer to the set of connectors on the motherboard that link to the buttons and indicators on the front of the computer case—like the power switch, reset switch, power LED, HDD LED, and sometimes audio or USB ports.

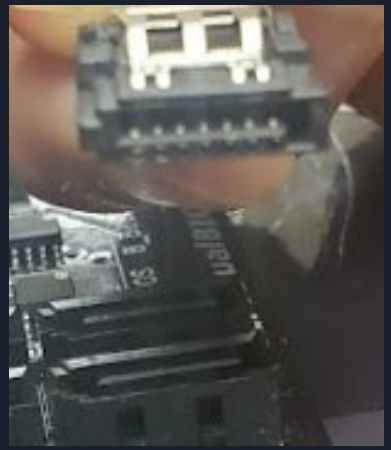






Connector Keying

Connector keying is a design that only allows a component to go in one way, preventing short circuits and broken components.



In this example, the cable and connector are both L-shaped in a way that only one orientation fits.

In another example, the RAM and the connector have a notch that only allow one orientation when installing



Case

A PC case's primary functions are to physically protect internal components from damage and the environment, provide a standard structure for mounting and organizing parts like the motherboard and drives, facilitate cooling and airflow, house front-panel I/O for user convenience, and help with cable management.

